Activity 1: The First Steps of a Digital Investigation

Alignment

Module 1 – Introduction to Computer Forensics and Investigations

Module 2 – Corporate Investigation Procedures

Nelson 7e Modules: Module 1, Module 2, Module 5

Free Resources: NIST SP 800-86, UNODC Guidelines

Learning Objectives

Describe the steps of a digital forensic investigation process.

Apply chain-of-custody principles to a provided scenario.

Document evidence handling using a professional-style evidence log.

- Explain why proper procedures matter for admissibility in court.

Estimated Time

1.5 to 2.5 hours

Scenario

You are a junior digital forensic investigator in a corporate environment. A manager has discovered suspicious files on an employee's workstation. You are tasked with:

- Identifying the first response actions you should take.
- Documenting the chain of custody for a seized USB drive and workstation.

- Preparing a short written briefing explaining your actions and how they align with accepted forensic standards (NIST SP 800-86, UNODC guidelines).

Instruction

Read the scenario.

Step 1 – Identify first response actions: In ½ to 1 page, list what you would do immediately (secure the scene, preserve evidence, avoid contamination, etc.).

Step 2 – Chain of custody form: Complete a one-page evidence log (template provided by instructor). Record details such as description, serial numbers, evidence ID, collection date/time, and who handled it.

Step 3 – Written briefing: In 1–2 pages, explain your reasoning:

- Why chain of custody is critical.
- How your steps align with international best practices.
- What could go wrong if procedures are not followed.

Deliverables should be submitted as a single Word or PDF file.

Deliverables

Chain of Custody Log (completed template)

Written Briefing (1–2 pages, professional tone)

Activity 1: Written Briefing & Chain of Custody Template

Case Information

Complete the following Checklist before submitting Assignment 2.

Case Number: ______Employee-001-20260516________________________

Investigaor Name: _________xxg______________________

Date: _____________25 May 2026_________________

Written Briefing Template

1. Introduction

On 15 May, 2026, the manager discovered suspicious files on an employee's workstation. As a junior digital forensic investigator, I have been assigned to respond to this incident, preserve the digital evidence (i.e., seized USB drive and workstation) and document the chain of custody, adhering to NIST SP 800-86 and UNODC guidelines.

2. First Response Actions

As a Digital Evidence First Responder (DEFR), my training requires I arrive at an incident scene, assess the situation, and take precautions to acquire and preserve evidence. Therefore, the following actions were taken on my arrival at the scene:

i. Secure the workstation

This to ensure the original evidence is not tampered with or contaminated. This will include no further use such as shutting down, logging on any device in the workstation or generally interacting with the workstation. This will also include photographing the workstation screen, layout so as to have evidence of the visual record of the initial state. If the workstation is powered on, an acquisition of the volatile memory, using the necessary tool, is required before powering down. Then, the seizure of the workstation to preserve the original evidence.

ii. Seize the USB Drive

This is required to ensure the analysis of the drive in the forensic investigation lab. Evidence bag must be tamper-proof before removing the USB drive from the incident scene to avoid losing and contaminating the content of the drive.

iii. Document Evidence

This is required in order to ensure that the chain of custody is not compromised. This include details about the workstation and USB drive, including who recovered the evidence and when and who possessed it and when. This is called the chain-of-evidence form which helps document what has and has not been done with the original evidence and forensic copies of the evidence. This will include the description of the workstation and USB drive, vendor name, who recovered the evidence, date, and time etc.

3. Chain of Custody Documentation

A complete Chain of Custody log has been completed. The key details are summarized below:

i. EVT-001 or WS-001, which is Dell Latitude 5520 Laptop, with serial number DL5520-78452 and a USB drive, was collected Employee workstation – Corporate Office.

ii. Both items were recovered by Semiu Abiodun Okerayi (Digital Forensic Investigator) from Employee workstation – Corporate Office to the Digital Forensics Lab at 10:30hrs through an evidence bag for the USB drive and casing for the workstation.

iii. The log captures every person who handled the evidence, the date/time, and the purpose of transfer ensuring no compromise of the chain of custody.

4. Alignment with Standards

Explain how your actions align with at least two sources, such as NIST SP 800-86 and UNODC Digital Evidence Guidelines.

NIST SP 800-86:

i. According to this standard, it is required that a clearly defined chain of custody should be followed to avoid allegations of mishandling or tampering of evidence. This involves keeping a log of every person who had physical custody of the evidence, documenting the actions that they performed on the evidence and at what time, storing the evidence in a secure location when it is not being used, making a copy of the evidence and performing examination and analysis using only the copied evidence, and verifying the integrity of the original and copied evidence.

ii. Additionally, evidence should be photographed to provide visual reminders of the computer setup and peripheral devices.

UNODC:

i. The first responder identifies and protects the crime scene from contamination and preserves volatile evidence by isolating the users of all digital devices found at the crime scene.

ii. According to this guideline, documentation should include detailed information about the digital devices collected, including the operational state of the device - on, off, standby mode - and its physical characteristics, such as make, model, serial number, connections, and any markings or other damage.

iii. In addition to written notes, sketches, photographs and/or video recordings of the crime scene and evidence are also needed to document the scene and evidence.

5. Risks of Improper Handling

Discuss what could go wrong if procedures are not followed or the chain of custody is broken.

Many cases are thrown out because the chain of custody can't be proven or has been broken. This gives the possibility that the evidence has been compromised, and it is not legally admissible.

Lack of standardization and procedures can cause confusion and mistakes about the evidence collected. Therefore, the integrity of the evidence might not be guaranteed, and the evidence might not be reproducible, rendering it not legally admissible.

6. Conclusion

Following proper procedures help maintain consistent quality, reproducibility for all investigations, and prevents confusion and mistakes about evidence collected. This ensures evidence integrity and admissibility,

Reference:

i. Nelson, B., Phillips, A., & Steuart, C. (2019). Guide to Computer Forensics and Investigations, 6th Edition.

ii. National Institute of Standards and Technology. (2006). Guide to Integrating Forensic Techniques into Incident Response (NIST SP 800-86). U.S. Department of Commerce.

iii. United Nations Office on Drugs and Crime. (2014). Criminal Justice Handbook Series: Cybercrime and Electronic Evidence — A Guide for Judges. UNODC.

Activity 1: Chain of Custody Log

Case Information

Case Number: ___________Employee-001-20260515_____________________

Evidence ID: _________EVT-001 or WS-001_________________________

Item Description (make, model, serial number, etc.): _Dell Latitude 5520 Laptop, DL5520-78452____

Location Seized: _____Employee workstation – Corporate Office__________________________

Date/Time Collected: ________16-05-2026/10:30hrs____________________

Collected By (Name & Signature): _____xxg & ____

Transfer Log

Date/Time	From (Name & Signature)	To (Name & Signature)	Purpose of Transfer
16 May 2026 11:49hrs	xxg - Junior digital forensic investigator	Digital Forensics Lab Manager	Original evidence handover and copies in case of volatile memory for intake processing and analysis
16 May 2026 12:00hrs	Digital Forensics Lab Manager	Digital Forensics Staff - Evidence Storage Officer	Transfer to secure evidence locker to maintain the integrity of the evidence
16 May 2026 15:00hrs	Digital Forensics Staff - Evidence Storage Officer	Digital Forensics Analyst - Malware/Memory/Disk Forensics Analyst	Released for forensics analysis such as disk imaging and examination
20 May 2026 13:00hrs	Digital Forensics Analyst - Malware/Memory/Disk Forensics Analyst	Digital Forensics Staff - Evidence Storage Officer	Returned to the secured evidence locker initially assigned to the evidence after examination

22 May 2026 09:00hrs	Digital Forensics Staff - Evidence Storage Officer	xxg - Junior Digital Forensic Investigator	Investigation Report

Final Storage Location

________**Storage**____________Digital Forensics Lab / Evidence
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